JACC: HEART FAILURE VOL. 13, NO. 8, 2025
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ORIGINAL RESEARCH

MECHANICAL CIRCULATORY SUPPORT/CARDIAC TRANSPLANT

Patterns of Left Ventricular Regional Wall Motion Abnormalities After Brain Death and Their Clinical Significance

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ABSTRACT

BACKGROUND Left ventricular dysfunction is common in potential heart donors after brain death, but specific regional wall motion abnormality (RWMA) patterns in this population have not been well described.

OBJECTIVES This study aims to define and characterize RWMA patterns in potential heart donors after brain death by using a machine learning algorithm.

METHODS The Donor Heart Study enrolled 4,333 potential heart donors after brain death. All had a transthoracic echocardiogram (TTE) including RWMA assessment, with each segment analyzed for WMS (Wall Motion Score). Those with any RWMA (ie, any WMS >1) were classified using *k*-means clustering, and each cluster's associations with donor clinical characteristics, heart use, and recipient survival were assessed.

RESULTS The final analytical cohort included 913 initial TTEs. We identified 4 unique RWMA phenotypes: focal basal septal (FBS) (n = 500), basal and midventricular (n = 311), apical (n = 66), and global hypokinesis (n = 36). These phenotypes exhibited similar donor characteristics but differed in troponin, N-terminal pro-B-type natriuretic peptide levels (both lowest in FBS), and left ventricular ejection fraction (LVEF) (highest in FBS). On subsequent TTEs (performed in 314 donors with any RWMA), all phenotypes demonstrated significant improvement in LVEF. The FBS phenotype had the highest donor heart acceptance for transplantation (68%). Of the hearts accepted for transplantation, there were no significant differences by RWMA phenotype in recipient survival.

CONCLUSIONS Left ventricular dysfunction after brain death exhibits distinct RWMA phenotypes, which differ in terms of selected biomarkers and LVEF, but not in recipient survival of the hearts accepted for transplantation. (JACC Heart Fail. 2025;13:102511) © 2025 by the American College of Cardiology Foundation.

Left ventricular (LV) dysfunction is commonly observed after brain death in potential organ donors. Its underlying pathophysiology, prognostic importance, and why it is often reversible (in 58% of cases[1]) all remain poorly understood. A consequence of these knowledge gaps is that LV dysfunction is a leading reason (in up to 25% of cases) for donor heart nonacceptance for transplantation.[2,3]

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Manuscript received March 26, 2024; revised manuscript received March 10, 2025, accepted March 19, 2025.

ISSN 2213-1779/$36.00 https://doi.org/10.1016/j.jchf.2025.102511

ABBREVIATIONS AND ACRONYMS

CAD = coronary artery disease

FBS = focal basal septal

LV = left ventricle

LVEF = left ventricular ejection fraction

NT-proBNP = N-terminal pro-B-type natriuretic peptide

RWMA = regional wall motion abnormality

SMD = standardized mean difference

TTE = transthoracic echocardiogram

The end result is the discard of many potentially viable donor hearts, thereby delaying heart transplantation for thousands of critically ill patients on the waitlist.

Prior work suggests that this phenomenon primarily stems from the profound sympathetic nervous system activation that accompanies brain death,[4-6] resulting in myocardial calcium overload and myocardial cell injury.[7,8] This "neurocardiogenic" hypothesis is supported by the observation that regional wall motion abnormality (RWMA) patterns that occur after brain death do not typically match coronary artery distributions; rather, previously reported patterns include global hypokinesis and focal hypokinesis of the basal, midventricular, and apical segments, or a combination thereof.[9,10]

The relative prevalence of these patterns remains unknown, due to the small samples in which they have been described.[9-13] Moreover, the variety of these patterns suggests that multiple distinct mechanisms may be responsible. Given that sympathetic nerve terminals are mainly concentrated in the basal LV segments,[14,15] one would expect the "neurocardiogenic" mechanism to produce predominantly basal hypokinesis. However, this pattern is actually under-represented in prior (albeit small) studies of LV dysfunction.[9-13]

Filling these knowledge gaps regarding LV dysfunction in potential organ donors was a central aim of the DHS (Donor Heart Study).[1] Its size (n = 4,333 potential heart donors) and collection of serial transthoracic echocardiograms (TTEs) with core interpretation makes it uniquely suited to investigating RWMA patterns after brain death. In this analysis, we apply a machine learning algorithm to inform the classification of distinct phenotypes of RWMA after brain death. We further evaluate their prevalence, associated clinical characteristics, and prognostic significance—knowledge that could inform selection and avert discard of potentially viable donor hearts.

METHODS

STUDY OVERVIEW. As previously described, the DHS was a 5-year multicenter observational prospective cohort study that enrolled 4,333 potential heart donors after brain death.[1] It was coordinated at Stanford University and conducted at 8 organ procurement organizations across the United States. Adult donors (18-65 years of age) were eligible for study inclusion if they experienced brain death and tested negative for HIV. Those not considered for cardiac donation (eg, due to coroner restriction or known preexisting cardiac disease deemed to be a contraindication for transplantation) were excluded.

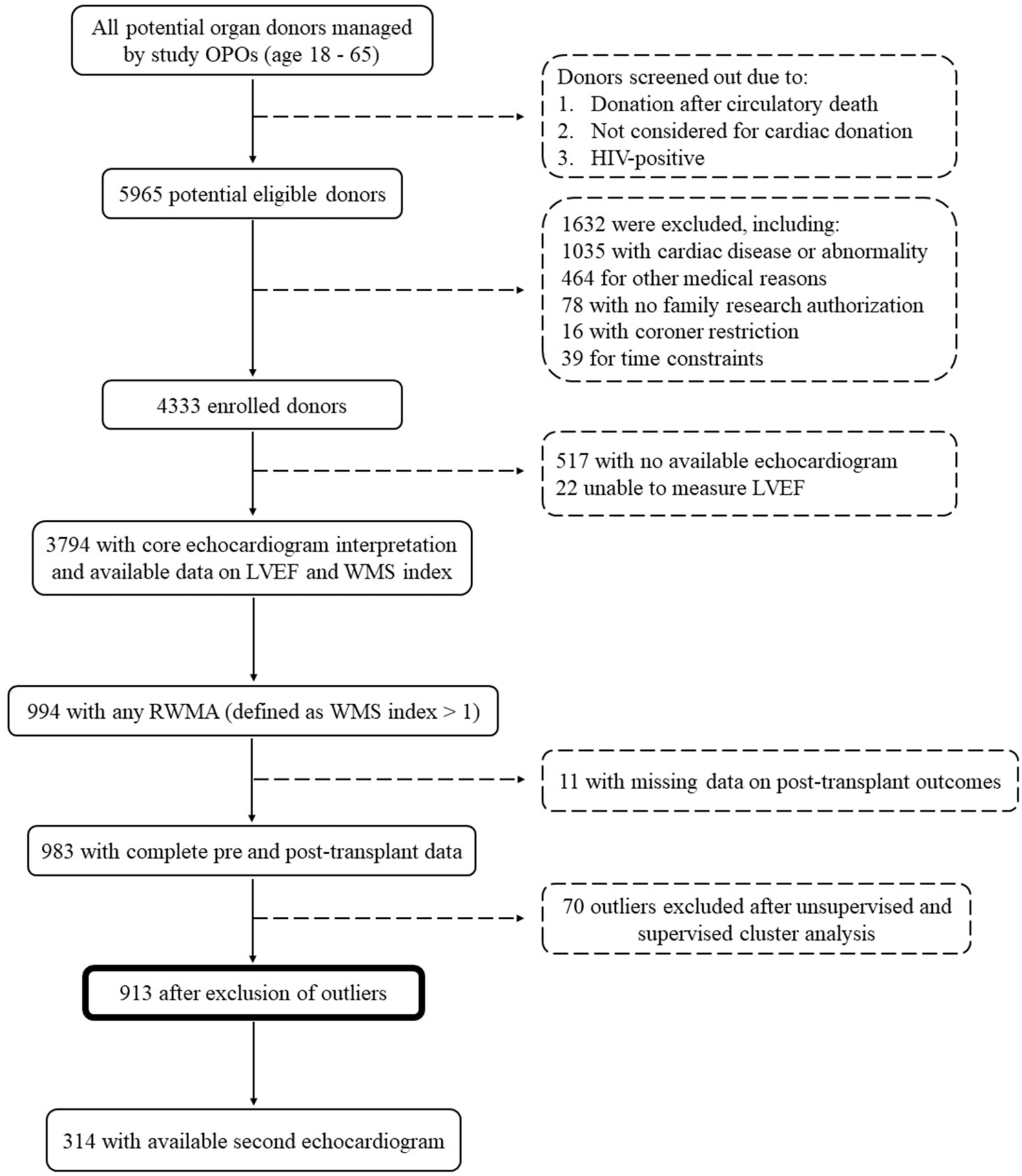
DATA AND TTE COLLECTION. Selected biomarkers were measured after declaration of brain death, authorization for donation and research, and subsequent achievement of hemodynamic stability. This time point, which was within 48 hours of declaration of brain death, roughly coincides with the start of donor management and is referred to as the "initial" time.

All enrolled donors had an initial TTE. If core interpretation identified LV dysfunction on initial TTE, defined as left ventricular ejection fraction (LVEF) $<$50%, a follow-up TTE was performed 24 $\pm$ 6 hours later. All TTEs included complete RWMA assessment, as quantified by the WMS (Wall Motion Score) index.[16] The WMS index is calculated by first assigning an integer value to each of 17 myocardial segments (1, normokinetic; 2, hypokinetic; 3, akinetic; and 4, dyskinetic) and then calculating the average across all 17 segments. A WMS index value $>$1 indicates the presence of 1 or more RWMAs (ie, at least 1 segment of the LV is hypokinetic, akinetic, or dyskinetic), with higher values indicating more severe abnormality. All TTEs were interpreted by an expert "core" reviewer (J.G.Z.), who was blinded to the local interpretation and all clinical data, including the timing and sequence of serial TTEs.

The principal investigators had full access to the data in the study and take responsibility for its integrity and the analyses presented herein. The study was approved by the Stanford University Institutional Review Board (protocol 31461), as well as by the research oversight committee at each participating organ procurement organization. The data that support the findings of this study are available from the corresponding author upon reasonable request.

STATISTICAL ANALYSES. Identifying RWMA phenotypes. To identify RWMA phenotypes, we used a combination of an unsupervised machine learning approach followed by reclassification based on clinical plausibility. First, within the subset of initial TTEs exhibiting any RWMA (ie, any WMS index $>$1), we applied the *k*-means clustering algorithm to classify distinct RWMA phenotypes. *K*-means clustering is an unsupervised machine learning technique that groups data points into clusters based on inherent patterns and similarities. Covariates in this analysis included each of the 17 distinct WMS measures (1 for each myocardial segment). To determine

FIGURE 1 DHS Consort Diagram, Demonstrating the Number of Eligible and Enrolled Donors, Availability of Echocardiographic and Clinical Data, and Exclusion of Outliers After Cluster Analyses



DHS = Donor Heart Study; LVEF = left ventricular ejection fraction; OPO = Organ Procurement Organization; RWMA = regional wall motion abnormality; WMS = Wall Motion Score.

the optimal number of clusters (k), we used the value of k that maximized the cubic clustering criterion. Statistical analyses were performed using SAS version 9.4 (SAS Institute).

The optimal number of clusters using this unsupervised approach was 8; the characteristics of each of these 8 preliminary clusters are shown in Supplemental Table 1. These 8 clusters were then consolidated (logic detailed in Supplemental Table 1) into 1 of 4 distinct phenotypes: 1) apical injury; 2) basal and midventricular injury; 3) global hypokinesis; and 4) focal basal septal (FBS) injury. Further examination of individual TTEs by phenotype revealed the presence of some outliers - TTEs with

TABLE 1 Donor Clinical Characteristics by RWMA Phenotype

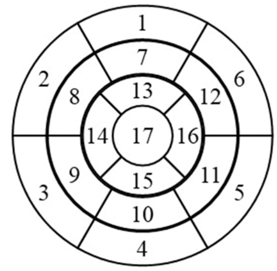
	Overall (N = 913)	Apical (n = 66, 7%)	Basal and Midventricular (n = 311, 34%)	Global Hypokinesis (n = 36, 4%)	Focal Basal Septal (n = 500, 55%)	SMD
Age, y	34.3 ± 11.7	38.7 ± 12.4	33.7 ± 11.5	35.4 ± 12.9	33.9 ± 11.5	0.225
Female	389 (42.6)	40 (60.6)	157 (50.5)	12 (33.3)	180 (36.0)	0.331
Race/ethnicity						0.327
White	736 (80.6)	58 (87.9)	264 (84.9)	25 (69.4)	389 (77.8)	
Black	145 (15.9)	6 (9.1)	34 (10.9)	9 (25.0)	96 (19.2)	
Asian	21 (2.3)	1 (1.5)	10 (3.2)	2 (5.6)	8 (1.6)	
Other	11 (1.2)	1 (1.5)	3 (1.0)	0 (0.0)	7 (1.4)	
BMI ≤20 (kg/m^2)	104 (11.4)	6 (9.1)	40 (12.9)	5 (13.9)	53 (10.6)	0.087
Medical history[a]						
Hypertension	167 (18.3)	15 (22.7)	61 (19.6)	9 (25.0)	82 (16.5)	0.119
Diabetes mellitus	55 (6.0)	6 (9.1)	16 (5.1)	6 (16.7)	27 (5.4)	0.213
Tobacco use	119 (13.2)	8 (12.5)	36 (11.7)	6 (16.7)	69 (14.0)	0.078
PHS risk criteria[b]	309 (33.8)	26 (39.4)	104 (33.4)	14 (38.9)	165 (33.0)	0.086
Cause of death						0.226
Head trauma	304 (33.3)	23 (34.8)	92 (29.6)	7 (19.4)	182 (36.4)	
CVA/stroke	206 (22.6)	13 (19.7)	79 (25.4)	9 (25.0)	105 (21.0)	
Anoxia and other	403 (44.1)	30 (45.4)	140 (45.0)	20 (55.6)	213 (42.6)	
Baseline characteristics[a]						
Temperature (°C)	36.8 ± 0.9	37.0 ± 0.8	36.9 ± 1.0	36.7 ± 1.2	36.8 ± 0.8	0.171
Mean arterial pressure, mm Hg	88.0 ± 16.9	92.7 ± 18.9	88.7 ± 18.0	91.1 ± 17.9	86.6 ± 15.6	0.196
Serum sodium, mmol/L	146.9 ± 7.9	146.5 ± 8.1	145.9 ± 7.8	145.8 ± 6.6	147.6 ± 7.9	0.081
BUN/Cr ratio	20.2 ± 13.2	19.1 ± 13.0	21.4 ± 14.1	19.5 ± 12.5	19.7 ± 12.7	0.090
Arterial pH	7.4 ± 0.1	7.3 ± 0.1	7.4 ± 0.1	7.4 ± 0.1	7.4 ± 0.1	0.176
Lactate, mmol/L	4.00 ± 3.43	4.10 ± 2.19	3.87 ± 3.10	4.74 ± 3.53	4.04 ± 3.80	0.137
NT-proBNP, pg/mL	4,318.95 ± 8,669.31	7,937.00 ± 12,543.81	5,856.10 ± 10,141.32	4,589.80 ± 6,364.35	2,608.82 ± 6,271.61	0.317
Troponin-I, ng/mL	3.6 ± 27.2	2.7 ± 4.4	5.5 ± 33.7	2.1 ± 3.3	2.7 ± 25.1	0.089
Troponin-I, ng/mL						0.415
T1 (≤0.06)	194 (22.2)	10 (15.6)	42 (14.2)	2 (6.2)	140 (29.0)	
T2 (0.06-0.42)	243 (27.8)	12 (18.8)	82 (27.7)	7 (21.9)	142 (29.5)	
T3 (>0.42)	437 (50.0)	42 (65.6)	172 (58.1)	23 (71.9)	200 (41.5)	
Available second TTE	314 (34.4)	28 (42.4)	162 (52.1)	16 (44.4)	108 (21.6)	0.336
Available coronary angiography	278 (30.4)	17 (25.8)	77 (24.8)	7 (19.4)	177 (35.4)	0.185
Coronary angiography[c]						0.377
Normal	233 (25.5)	10 (15.2)	65 (20.9)	7 (19.4)	151 (30.2)	
Minor CAD	27 (3.0)	6 (9.1)	8 (2.6)	0 (0.0)	13 (2.6)	
Major CAD	17 (1.9)	1 (1.5)	3 (1.0)	0 (0.0)	13 (2.6)	
Abnormal NOS	1 (0.1)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)	
Not available	635 (69.6)	49 (74.2)	234 (75.2)	29 (80.6)	323 (64.6)	
Administered therapies						
Steroids	691 (75.7)	46 (69.7)	241 (77.5)	29 (80.6)	375 (75.0)	0.136
Thyroxine	183 (20.0)	13 (19.7)	81 (26.0)	9 (25.0)	80 (16.0)	0.145
Number of vasopressors[d]	0.62 ± 0.51	0.74 ± 0.53	0.65 ± 0.51	0.82 ± 0.67	0.57 ± 0.49	0.253
Outcomes						
Donor heart accepted	498 (54.5)	13 (19.7)	132 (42.4)	12 (33.3)	341 (68.2)	0.568
Organ accept rank[e]	4 (2-16)	5 (3-19)	6 (3-22)	4 (2-10)	4 (2-13)	0.284
Recipient 2-y survival[f]	411 (82.6)	10 (83.3)	112 (89.6)	10 (83.3)	279 (85.6)	0.100

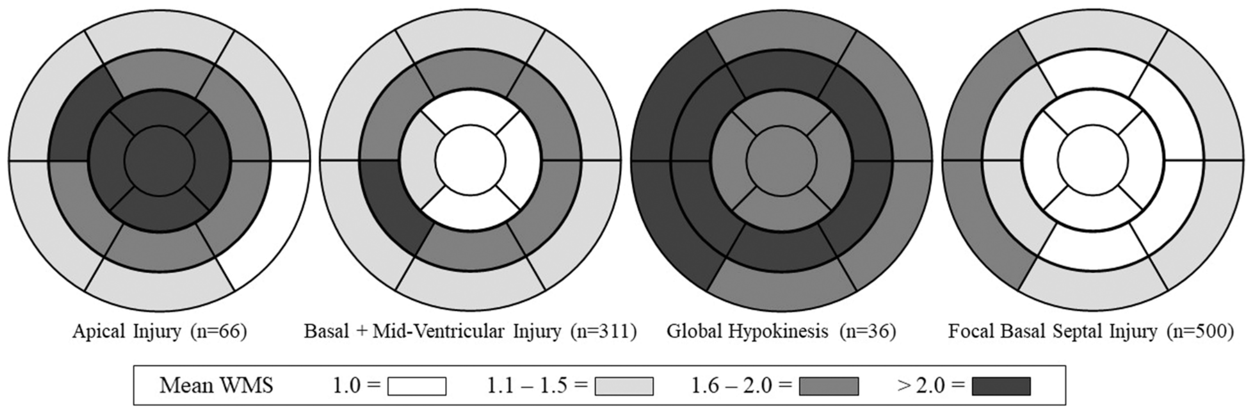
Values are mean ± SD, n (%), or median (Q1-Q3). [a]Proportions were calculated based on nonmissing variables. Missing values were <1% in the following variables: hypertension, diabetes mellitus, tobacco use, temperature, mean arterial pressure, serum sodium, BUN/Cr ratio, troponin-I, and arterial pH. [b]Refers to PHS designation of donors at increased risk of transmitting hepatitis B, hepatitis C, and HIV. [c]"Major CAD" refers to the presence of ≥50% stenosis in any major epicardial artery. "Minor CAD" indicates an abnormal coronary angiogram with no such stenoses. "Abnormal NOS" indicates an abnormal coronary angiogram but the degree of stenosis was not otherwise specified. [d]Includes dopamine, dobutamine, phenylephrine, epinephrine, norepinephrine, and vasopressin. [e]"Organ Accept Rank" refers to how many transplantation centers rejected the donor heart before it was accepted (ie, "4" indicates the 4th transplantation center accepted the donor heart). [f]The SMD for recipient 2-year survival was calculated based on nonmissing survival data, which comprised <2% of the total cohort.

BMI = body mass index; BUN = blood urea nitrogen; CAD = coronary artery disease; Cr = creatinine; CVA = cerebrovascular accident; LVEF = left ventricular ejection fraction; NOS = not otherwise specified; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PHS = Public Health Service; SMD = standardized mean difference; T = tertile; TTE = transthoracic echocardiogram; WMSI = Wall Motion Score index.

FIGURE 2 **RWMA Phenotypes in Donor Hearts**

Basal Level	Mid-Ventricular Level	Apical Level
1. Basal Anterior	7. Mid Anterior	13. Apical Anterior
2. Basal Anteroseptal	8. Mid Anteroseptal	14. Apical Septal
3. Basal Inferoseptal	9. Mid Inferoseptal	15. Apical Inferior
4. Basal Inferior	10. Mid Inferior	16. Apical Lateral
5. Basal Inferolateral	11. Mid Inferolateral	17. Apex
6. Basal Anterolateral	12. Mid Anterolateral	





RWMAs are depicted using the 17-segment model of the LV. The WMS is defined as follows: 1 = normokinetic, 2 = hypokinetic, 3 = akinetic, and 4 = dyskinetic. We identified 4 unique RWMA phenotypes: apical injury, basal and midventricular injury, global hypokinesis, and FBS injury. The FBS phenotype was the most prevalent phenotype. FBS = focal basal septal; LV = left ventricle; other abbreviations as in Figure 1.

patterns of RWMA that deviated significantly from their assigned phenotype. Thus, we identified and excluded 70 outliers as detailed in Supplemental Methods. The wall motion assessments of these 70 outliers are shown in Supplemental Table 2.

Clinical and prognostic associations by RWMA phenotype. Donor covariates spanned multiple domains, including demographics, past medical and social history, hemodynamic values, medications administered, and laboratory test results. Most covariates were expressed as categorical variables, with thresholds based on use in comparable previous studies. Troponin-I was expressed as a categorical variable in tertiles. Lactate and N-terminal pro-B-type natriuretic peptide (NT-proBNP) were expressed as continuous variables. The standardized mean difference (SMD) was used to quantify and compare the associations between covariates (both categorical and continuous) and RWMA phenotypes. The SMD was interpreted according to Cohen's d, which is as follows: SMD >0.2 indicates a small difference, SMD >0.5 indicates a substantial moderate difference, and SMD >0.8 indicates a substantial large difference.[17]

We depicted LVEF and WMS index trends between initial and follow-up TTE using box and whisker plots. The trends were analyzed using paired Student's *t*-tests and 2-sided values of $P < 0.05$ were considered statistically significant. The associations between RWMA phenotypes and post-transplantation recipient 2-year survival were assessed using Kaplan-Meier curves and log-rank tests.

RESULTS

STUDY POPULATION. A total of 4,333 potential donors were enrolled (Figure 1). Initial TTEs of adequate quality for complete RWMA assessment were obtained in 3,794 (87.5%) enrolled potential donors. Of these initial TTEs, approximately one fourth had RWMA (n = 994). After exclusion of outliers, the final analytical cohort included 913 initial TTEs with complete pretransplantation and post-transplantation data. There were no meaningful clinical differences between the final analytical cohort and the outliers (Table 1, Supplemental Table 3). Donors with RWMA tended to be younger

TABLE 2 Donor Initial Left Ventricular Function by Regional Wall Motion Abnormality Phenotype

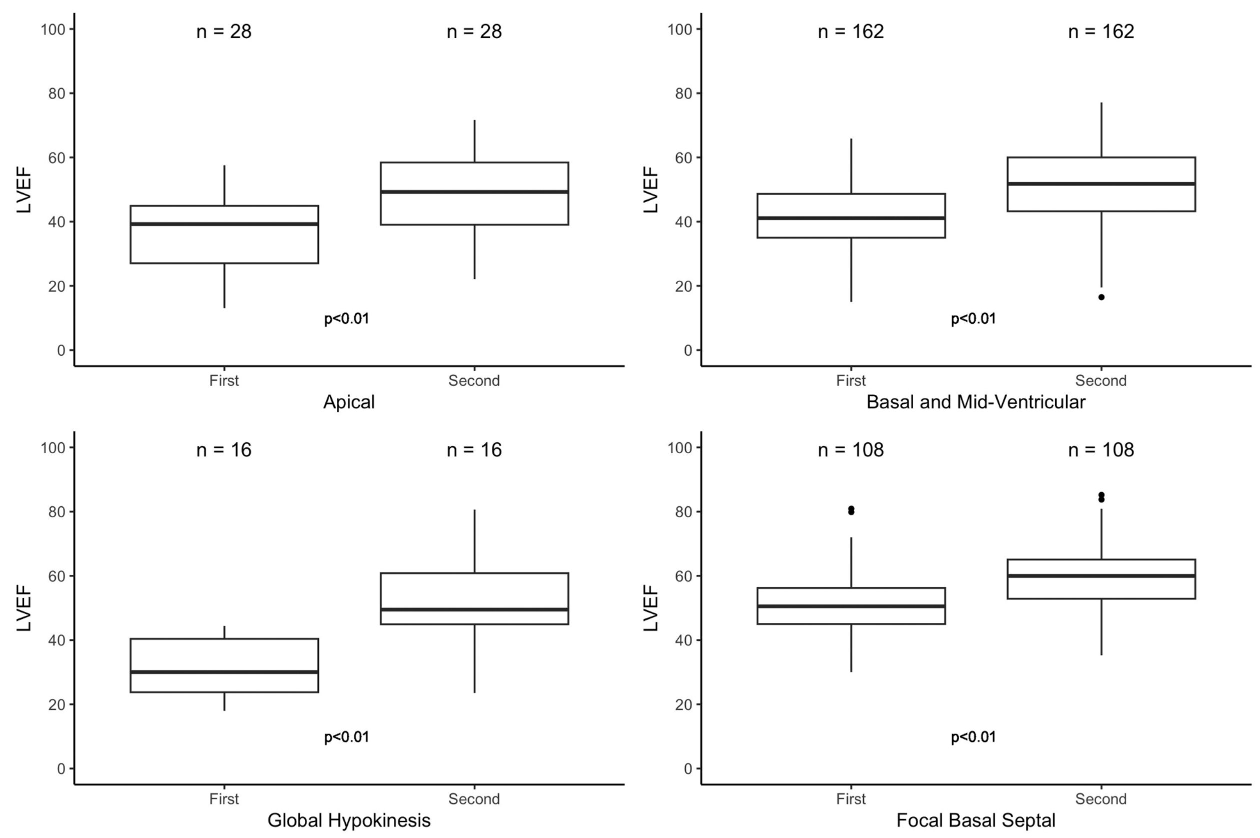
	Overall	Apical	Basal and Midventricular	Global Hypokinesis	Focal Basal Septal	SMD
Initial LVEF, %	48.2 ± 12.5	38.6 ± 14.7	42.5 ± 10.9	30.1 ± 10.2	54.3 ± 9.1	1.185
Initial LVEF						1.502
<35%	115 (12.6)	24 (36.4)	67 (21.5)	22 (61.1)	2 (0.4)	
35%-49%	342 (37.4)	29 (43.9)	162 (52.1)	14 (38.9)	137 (27.4)	
≥50%	456 (49.9)	13 (19.7)	82 (26.4)	0 (0.0)	361 (72.2)	
Initial WMSI	1.31 (1.14-1.53)	1.65 (1.47-1.94)	1.53 (1.41-1.69)	2.00 (1.94-2.06)	1.18 (1.12-1.24)	2.416
Initial WMSI >1.5	253 (27.7)	47 (71.2)	169 (54.3)	36 (100.0)	1 (0.2)	6.313

Values are mean ± SD, n (%), or median (Q1-Q3).
Abbreviations as in Table 1.

and had lower rates of traditional coronary artery disease (CAD) risk factors like hypertension, diabetes mellitus, and tobacco use (Table 1). Of those with available angiogram data, a minority had CAD (17.6%) and a majority of those were classified as nonobstructive CAD (55.5%) (Table 1). The most common cause of brain death was cerebral anoxia.

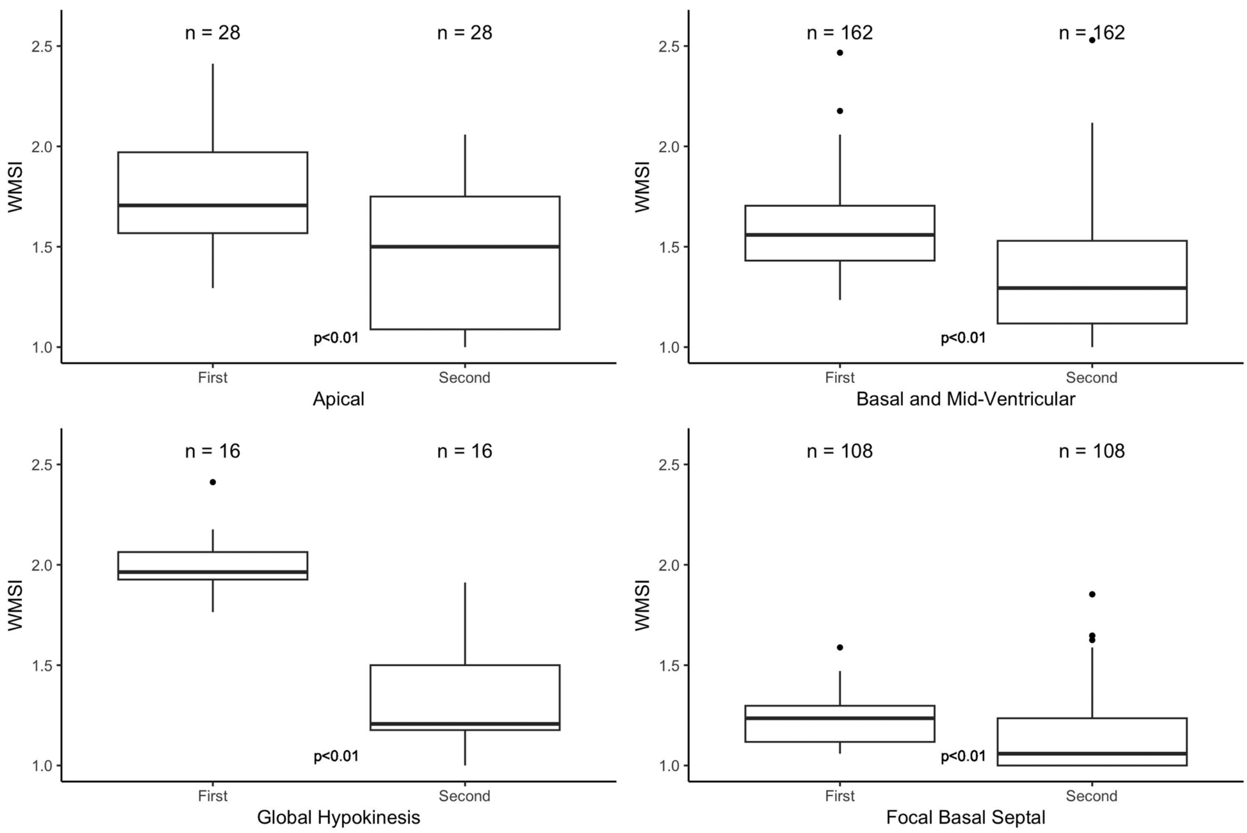
IDENTIFICATION OF RWMA PHENOTYPES. We identified 4 unique RWMA phenotypes (Figure 2):

FIGURE 3 LVEF on Initial Echocardiogram and Second Echocardiogram



LVEF by RWMA phenotype is depicted with a standard box and whisker plot. The horizontal line represents the median, the "box" encompasses the first quartile to the third quartile, the "whiskers" encompasses the range, and the bullet points represent data outliers (any data points that fall below quartile 1-1.5 × IQR or above quartile 3 + 1.5 × IQR). Statistical significance was assessed using paired Student's *t*-tests, and changes were deemed significant if $P < 0.05$. If core interpretation identified LV dysfunction on initial echocardiogram, defined as LVEF <50%, a second echocardiogram was performed 24 ± 6 hours later. All RWMA phenotypes had improvements in LVEF on second echocardiogram. Donors with global hypokinesis demonstrated the greatest improvement in LVEF. Abbreviations as in Figures 1 and 2.

FIGURE 4 WMS Index on Initial Echocardiogram and Second Echocardiogram



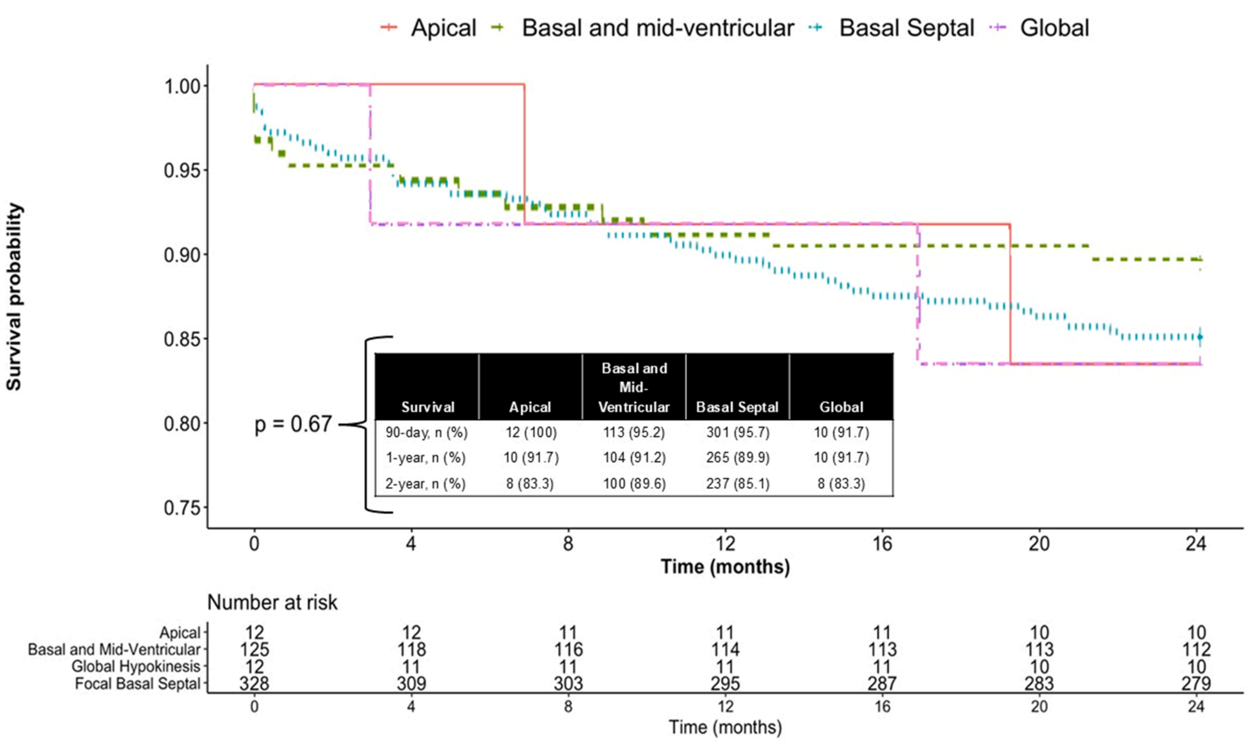
WMS index by RWMA phenotype is depicted with a standard box and whisker plot. The horizontal line represents the median, the "box" encompasses the first quartile to the third quartile, the "whiskers" encompasses the range, and the bullet points represent data outliers (any data points that fall below quartile 1-1.5 × IQR or above quartile 3 + 1.5 × IQR). Statistical significance was assessed using paired Student's *t*-tests and changes were deemed significant if $P < 0.05$. If core interpretation identified LV dysfunction on initial echocardiogram, defined as LVEF <50%, a second echocardiogram was performed 24 ± 6 hours later. All RWMA phenotypes had improvements in WMS index on second echocardiogram. Donors with global hypokinesis demonstrated the greatest improvement in WMS index. Abbreviations as in Figures 1 and 2.

apical injury, basal and midventricular injury, global hypokinesis, and FBS injury. The FBS phenotype was the most prevalent (n = 500) (55%) followed by basal and midventricular (n = 311) (34%), apical (n = 66) (7%), and global hypokinesis (n = 36) (4%).

ASSOCIATIONS OF RWMA PHENOTYPES WITH CLINICAL CHARACTERISTICS. The different RWMA phenotypes largely exhibited similar donor characteristics but had a few key differences (Table 1). There were differences in LVEF, which ranged from a mean of 30.1% in the global hypokinesis phenotype to 54.3% in the FBS phenotype. The FBS phenotype also had the lowest WMS index (median: 1.18 [Q1-Q3: 1.12-1.24]) and the global hypokinesis phenotype had the highest WMS index (median: 2.00 [Q1-Q3: 1.94-2.06]) (Table 2). There were also differences in troponin-I and NT-proBNP, which were both lowest in the FBS phenotype. There were 314 follow-up TTEs available, all of which were obtained within 24 ± 6 hours of the initial TTE per trial protocol. Of those with paired data, all phenotypes demonstrated significant improvement in LVEF and WMS index (Figures 3 and 4).

DONOR HEART USE AND RECIPIENT SURVIVAL BY RWMA PHENOTYPES. The FBS phenotype had the highest donor heart use for transplantation (68%). Of hearts accepted for transplantation, there were no significant differences in recipient 2-year survival by RWMA phenotype (Table 1, Figure 5). Donor hearts with less severe injury (WMS index ≤1.5) generally had higher rates of donor heart use than those with more severe injury (WMS index >1.5), but had similar

FIGURE 5 Kaplan-Meier Curves for Recipient 2-Year Survival, Stratified by the 4 RWMA Phenotypes



The 4 RWMA phenotypes did not differ in terms of recipient 2-year survival. Statistical significance was assessed using log-rank tests and differences were deemed significant if $P < 0.05$. Abbreviations as in Figure 2.

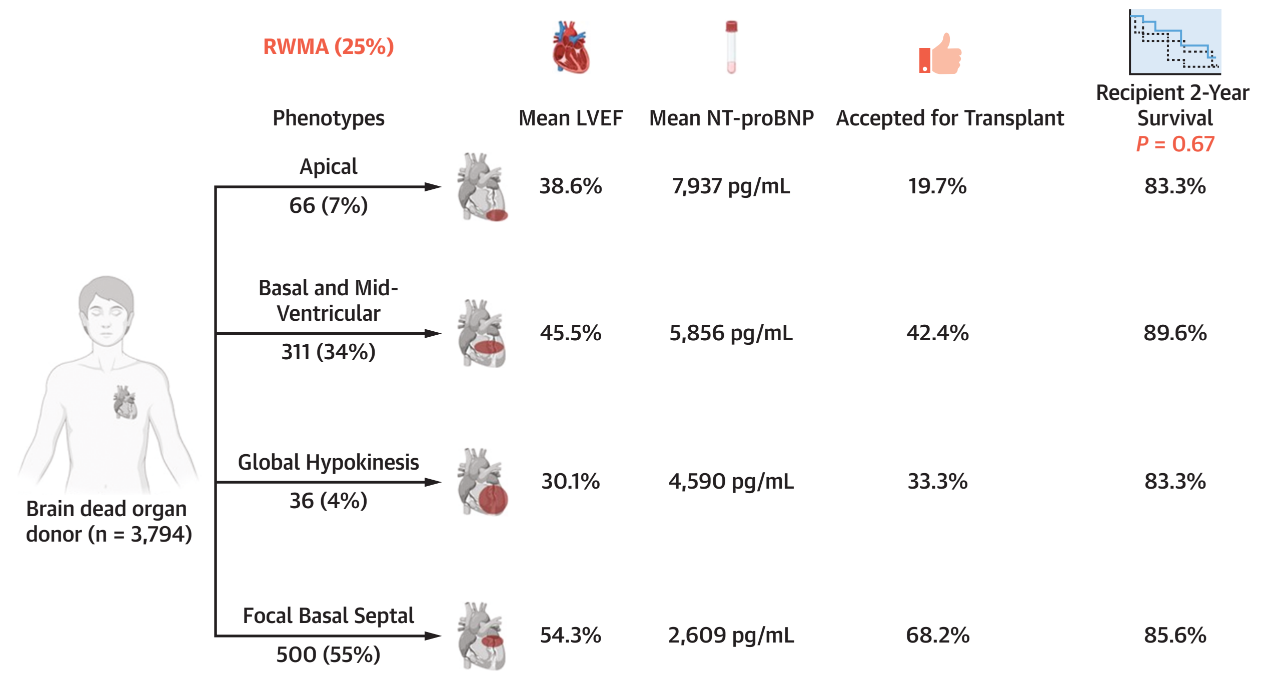
recipient 2-year survival for all RWMA phenotypes (Supplemental Table 4). The presence or absence of a repeat TTE did not alter rates of donor heart use or recipient 2-year survival for all RWMA phenotypes (Supplemental Table 5).

DISCUSSION

Using a combined supervised and unsupervised machine learning approach, we identified 4 unique RWMA phenotypes in brain-dead organ donors: apical injury, basal and midventricular injury, global hypokinesis, and FBS injury (Central Illustration). The FBS phenotype was the predominant phenotype and accounted for more than half of the cohort. Compared with donors without RWMA,[1] donors with the FBS phenotype had higher levels of troponin-I and NT-proBNP. Compared with other donors with RWMA, donors with the FBS phenotype had the highest initial LVEF, lowest troponin-I and NT-proBNP levels, and highest donor heart acceptance for transplantation. Of those with paired data, all RWMA phenotypes demonstrated improvements in LVEF and WMS index during the donor management period. Of the hearts accepted for transplantation, there were no differences in recipient 2-year survival by RWMA phenotype.

As the predominant form of LV dysfunction in donor hearts after brain death, the FBS phenotype offers several insights and raises several questions. The putative mechanisms of LV dysfunction after brain death stems from the “neurocardiogenic” hypothesis—brain death triggers excessive catecholamine release from myocardial sympathetic nerve terminals, which lead to myocyte calcium overload and cellular injury.[4-8] Because sympathetic nerve terminals are mostly concentrated in the basal LV segments,[14,15] basilar injury is expected to be the

CENTRAL ILLUSTRATION RWMA Phenotypes in Donor Hearts and Their Associated Clinical Characteristics and Prognostic Significance



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We identified 4 unique LV RWMA phenotypes in donors after brain death: apical injury, basal and midventricular injury, global hypokinesis, and FBS injury. These phenotypes differ in terms of initial LVEF, NT-proBNP levels, and donor heart acceptance rate, but not in recipient 2-year survival. FBS = focal basal septal; LV = left ventricular; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; RWMA = regional wall motion abnormality; WMS = Wall Motion Score.

predominant form of myocardial injury, and as a form of basilar injury, the FBS phenotype provides evidence for the "neurocardiogenic" hypothesis.

It is unclear why the basal septum is selectively affected in the majority of these patients. One possibility is there are variations in the distribution of sympathetic innervation within the basal segments themselves. One experimental study demonstrated that, of the basal segments in primate hearts, the septum had the highest density of catecholamines.[14] Another possible explanation is that the FBS phenotype represents a minor form of basilar injury as it was associated with the lowest levels of troponin-I and NT-proBNP and highest initial LVEF. Conceivably, the profound autonomic dysregulation that follows catastrophic brain injury disrupts the balance and distribution of sympathetic and parasympathetic myocardial innervation in the basal LV segments, leading to varying degrees of basal LV myocardial injury. Further research investigating this autonomic dysregulation would further elucidate the pathophysiological mechanism behind LV dysfunction in donor hearts after brain death.

In addition to the FBS phenotype, we identified alternative phenotypes of LV dysfunction in donor hearts after brain death including apical injury, basal and midventricular injury, and global hypokinesis. Basal and midventricular injury was much more prevalent (~35%) than apical injury (~7%) and global hypokinesis (~4%). These findings are consistent with prior descriptions of RWMA patterns after brain death.[9-13] The presence of these alternative, less common patterns could be related to inherent patient differences in myocardial sympathetic nerve terminal distribution, where certain patients may have greater relative concentrations of sympathetic nerve terminals in the apical segments when compared with the basal segments. Additionally, there are likely variations in myocardial cell responsiveness to adrenergic stimuli. Several experimental studies found that the apical myocardium has greater beta-adrenergic receptor density and increased myocardial responsiveness to adrenergic stimulation, despite its sparse sympathetic innervation.[18,19]

Overall, we posit these RWMA phenotypes provide additional evidence for "neurocardiogenic" myocardial injury because basilar and midventricular injury predominated, many of the RWMA patterns do not match coronary artery distributions, and LV function was highly reversible during a short period of time. Our findings argue against direct myocardial ischemia as a pathologic mechanism for several reasons. First, donors had lower rates of traditional CAD risk factors like hypertension, diabetes mellitus, and tobacco use when compared with contemporary studies of patients who had a myocardial infarction.[20] Additionally, of the donor hearts with available angiogram data, a minority (17.6%) had CAD and more than half of those had nonobstructive CAD (55.5%) (**Table 1**). Although it is possible there may be an element of coronary vasospasm, a prior study of patients with stress cardiomyopathy, a similar clinical entity to neurogenic myocardial injury, demonstrated a minority actually had inducible coronary vasospasm.[21] Additionally, prior experimental studies have demonstrated that cardiac myofibrillar degeneration in patients who die from acute stroke is histologically identical to cardiac lesions of catecholamine infusion and occurs in the vicinity of cardiac nerves and not in the macrovascular distribution seen in patients with CAD.[22] The lesions also differ from the necrosis seen in CAD because they are visible within minutes of onset, and the cells die in a hypercontracted state with contraction bands, associated mononuclear infiltration, and early calcification.[23]

However, the "neurocardiogenic" hypothesis likely incompletely captures the full spectrum of catecholamine-mediated injury because variations in nerve terminal distribution, receptor distribution, and receptor responsiveness likely lead to the observed variation in RWMA pattern. There may also be an element of inflammation unrelated to sympathetic nervous system activation. Brain death initiates a profound inflammatory response that is primarily driven by the release of proinflammatory cytokines and disruption to the blood brain barrier, the combination of which exposes cardiomyocytes to the deleterious effects of both systemic and neurogenic inflammatory pathways.[24] Prior studies have found that brain death-associated cytokines such as tumor necrosis factor-α and interleukin-6 can depress myocardial contractility via disruption of intracellular calcium homeostasis and induce ventricular myocyte apoptosis.[25,26] However, it is unclear if specific inflammatory pathways are associated with specific RWMA phenotypes and this should be explored in future studies.

Among the subset of potential heart donors with paired data (follow-up TTE within 24 $\pm$ 6 hours), we found all RWMA phenotypes demonstrated improvements in LVEF and WMS index. The degree of improvement was lowest in the FBS phenotype and highest in the global hypokinesis phenotype, likely due to having the best and worst initial LVEF and WMS index, respectively. As expected, donors with the FBS phenotype had the highest heart acceptance for transplantation, but ultimately there were no differences in 2-year survival between recipients of donor hearts with distinct RWMA phenotypes.

Based on our findings, the presence or lack of a certain RWMA phenotype does not have meaningful clinical implications because recipient survival in the early post-transplantation course was similar across all phenotypes. This likely indicates that LV dysfunction and RWMA have less of an impact on post-transplantation outcomes when compared with other established risk predictors like donor age, donor/recipient predicted heart mass mismatch, or donor-transmitted CAD.[27] However, certain RWMA phenotypes could inform treatment protocols during donor management. If myocardial injury after brain death is truly catecholamine-mediated, it is conceivable that the mitigation of this process with adrenergic blockage could be beneficial, particularly early in donor management or in RWMA phenotypes with evidence of more extensive neurocardiogenic myocardial injury like global hypokinesis. A prior experimental study of brain-dead pigs found the use of labetalol immediately before and after the induction of brain death led to preservation of myocardial contractile function.[28] A subsequent single-center retrospective study found the use of adrenergic blocking agents like esmolol and urapidil (an α_1-adrenergic blocker) after brain death was associated with preserved LV function.[29] However, these are far from definitive conclusions given the studies' limitations. Furthermore, it is not feasible to treat patients with adrenergic blockade in the moments leading up to brain death, and the use of adrenergic blocking agents after brain death would be problematic given the hemodynamic issues that typically arise like vasodilation and hypotension. Much more attention has been given to vasopressor and hormonal therapies like vasopressin, corticosteroids, and thyroid hormone, which are common components of donor management protocols.[30] In particular, combination hormonal therapy with vasopressin, corticosteroids, and thyroid hormone, as compared with no or partial hormone replacement, has been associated with improved recipient 1-year survival and reduced incidence of early cardiac allograft dysfunction.[31] It appears this effect was driven by the use of corticosteroids, which suggests that mitigation of the systemic inflammation after brain death can preserve LV function and improve clinical outcomes. Future research should explore if specific RWMA phenotypes are more due to systemic inflammation vs sympathetic nervous system activation, and thus have more to gain from anti-inflammatory interventions.

STUDY STRENGTHS AND LIMITATIONS. Several strengths of our study merit mention. First, the DHS is the largest and most comprehensive cohort of potential heart donors to date. Second, the prospective and multicenter design limits bias and enhances generalizability. Third, all TTEs had core interpretation with a standardized protocol by a single blinded expert reviewer. Local interpretation of TTEs introduces inconsistences in LV assessment, as evidenced by prior reports of significant discordance between local and core interpretation of donor TTEs.[32] Finally, the use of serial echocardiography enabled us to characterize reversibility of LV dysfunction.

Several limitations of the study should also be acknowledged. First, we excluded 70 outliers from our analytic cohort because their patterns of RWMA deviated significantly from their assigned phenotype. Although we recognize that exclusion of data can lead to misinterpretation of results and introduce bias, the impact was likely minimal as the outliers made up a small portion of the sample size (<10%). Furthermore, when comparing analyses with and without the inclusion of outliers, there were no meaningful changes in the SMDs (**Table 1**, Supplemental Table 3). Second, the supervised portion of our *k*-means cluster analysis introduced an element of subjectivity that can limit reproducibility and generalizability. To our knowledge, there are no established standards in categorizing RWMA phenotypes using a combined unsupervised and supervised *k*-means cluster analysis. Thus, our approach was to reassign clusters according to previously described RWMA phenotypes and clinical plausibility (Supplemental Table 1).[9-13] Third, our study excluded donors with known preexisting LV dysfunction, so our results cannot be applied to those patients. Fourth, because our study was not a central analysis, biomarkers were not prospectively obtained and analyzed in a core laboratory. As different assays were used, our biomarker data are heterogenous and, therefore, limit the conclusions we can draw from them. Finally, there were a limited number of follow-up TTEs available for analysis for a variety of reasons. As donor management occurs around the clock, it was not feasible to perform core interpretation in real time. Thus, the decision to perform a follow-up TTE depended on a local cardiologist's interpretation of the initial TTE, and specifically (as per study protocol) whether the local read reported an LVEF <50%. The local and core interpretations were sometimes discordant, such that 165 follow-up TTEs (34.8%) were not obtained in donors with LVEF <50% as the local read reported an initial LVEF ≥50% (Supplemental Table 6). As a result, several donor hearts were excluded from the paired analyses, which limits the statistical power of these data. Additionally, some donor hearts with initial LVEF <50% were deemed ineligible for cardiac donation, so a follow-up TTE was deferred if it would have delayed the procurement of other organs. Collectively, our estimates of LVEF and WMS index improvement are based on a subset of 45% of all potential donors with initial LV dysfunction. However, a previous comparison of clinical characteristics between potential donors with initial LV dysfunction who did and did not have a repeat TTE demonstrated no measurable differences.[1] Additionally, there were no significant differences in rates of donor heart use and recipient 2-year survival between those who did and did not have a repeat TTE (Supplemental Table 5).

CONCLUSIONS

We used a machine learning algorithm to identify previously reported patterns of LV dysfunction after brain death as well as a unique and highly prevalent FBS phenotype, which appears to represent a form of "neurocardiogenic" myocardial injury. Many of these RWMA patterns are not in distributions typical of CAD and are highly reversible. For RWMA patterns that are likely to be neurocardiogenic (ie, FBS), there may not be need for a coronary angiogram or a repeat echocardiogram. Most importantly, although these RWMA phenotypes differed in terms of selected biomarkers, initial LVEF, and donor heart acceptance for transplantation, recipients of hearts that were transplanted had similar survival. This study provides further evidence that the presence of LV systolic dysfunction on an initial echocardiogram should not, by itself, exclude a donor heart from consideration for transplantation. We hope the results of our study will inform increased use of donor hearts with RWMA, which would help alleviate donor organ scarcity and its dire consequences for patients awaiting heart transplantation.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This study was funded by the National Institutes of Health (NIH)/National Heart, Lung, and Blood Institute (R01 HL125303). The Stanford REDCap platform (http://redcap.stanford.edu) was developed and operated by the Stanford Medicine Research Information Technology team. The REDCap platform services at Stanford are subsidized by the Stanford School of Medicine Research Office, the National Center for Research Resources, and the National Center for Advancing Translational Sciences, NIH, through grant UL1 TR001085. The data reported here have been supplied by the Donor Management Goals Advisory Group. The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy of or interpretation by the Donor Management Goals Advisory Group. The authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: There were no differences in prognostic significance between 4 distinct RWMA patterns in donor hearts after brain death. The results of our study can inform increased use of donor hearts with RWMA, which would help alleviate donor organ scarcity and its dire consequences for patients awaiting heart transplantation.

TRANSLATIONAL OUTLOOK: Future studies are necessary to further elucidate mechanisms of "neurocardiogenic" injury because they could inform unique treatment protocols during donor management. Additionally, machine learning algorithms offer an objective and scalable data analytic tool for future use in large clinical trials.

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KEY WORDS donor heart, left ventricular dysfunction, machine learning, outcomes, regional wall motion abnormality

APPENDIX For an expanded Methods section as well as supplemental tables, please see the online version of this paper.